

Short Description of the Scene

The scene features a Utah teapot rendered with a custom GLSL shader that simulates a transmissive and reflective material within a meadow valley environment. The surrounding cubemap represents the valley and sky and is used both as the scene background and as an environment texture sampled by the shader using 6 images (back, front, top, bottom, right, left) . Reflection, refraction, Fresnel component, and chromatic dispersion allow the meadow valley to be seen distorted and reflected within the teapot, creating the illusion of a glass-like object interacting with its surroundings. Real-time controls enable live adjustment of material parameters to demonstrate their visual and physical effects. The implementation uses OpenGL with GLFW, GLAD, GLM, and ImGui, with all shader logic.